

Manual Testing & QA — Complete Syllabus

Zero → Job-Ready → Automation/SDET Roadmap | Topic Checklist

How to use: Har topic ko complete karne ke baad checkbox mark karo. Manual Testing ke topics pehle complete karo; baad mein SQL, API, Python, Selenium aur Automation Framework par move karo.

LEVEL 1 — Software Testing Fundamentals

No.	Topic	✓
1	Software kya hai?	☐
2	Software Testing kya hai?	☐
3	Testing ki need/importance	☐
4	Objectives of Testing	☐
5	Software Tester ka role	☐
6	QA kya hai?	☐
7	QC kya hai?	☐
8	QA vs QC vs Testing	☐
9	Error, Bug, Defect, Failure	☐
10	Verification vs Validation	☐
11	Tester ki responsibilities	☐
12	Testing ke principles	☐
13	Seven Principles of Software Testing	☐

LEVEL 2 — SDLC & STLC

No.	Topic	✓
14	SDLC kya hai?	☐
15	SDLC ke phases	☐
16	Waterfall Model	☐
17	V-Model	☐
18	Agile Model	☐
19	Iterative Model	☐
20	Spiral Model	☐
21	STLC kya hai?	☐
22	STLC ke phases	☐
23	Requirement Analysis	☐
24	Test Planning	☐
25	Test Case Development	☐
26	Test Environment Setup	☐
27	Test Execution	☐
28	Test Closure	☐

LEVEL 3 — Testing Levels

No.	Topic	✓
29	Unit Testing	■
30	Integration Testing	■
31	System Testing	■
32	Acceptance Testing	■
33	UAT — User Acceptance Testing	■
34	Alpha Testing	■
35	Beta Testing	■

LEVEL 4 — Testing Types

No.	Topic	✓
36	Functional Testing	■
37	Regression Testing	■
38	Retesting	■
39	Smoke Testing	■
40	Sanity Testing	■
41	Exploratory Testing	■
42	Ad-hoc Testing	■
43	End-to-End Testing	■
44	GUI Testing	■
45	Compatibility Testing	■
46	Performance Testing	■
47	Load Testing	■
48	Stress Testing	■
49	Volume Testing	■
50	Security Testing	■
51	Usability Testing	■
52	Reliability Testing	■
53	Scalability Testing	■
54	Accessibility Testing	■

LEVEL 5 — Testing Techniques

No.	Topic	✓
55	Black Box Testing	■
56	White Box Testing	■
57	Grey Box Testing	■
58	Equivalence Partitioning	■
59	Boundary Value Analysis	■
60	Decision Table Testing	■
61	State Transition Testing	■
62	Use Case Testing	■
63	Error Guessing	■
64	Pairwise Testing	■

LEVEL 6 — Test Documentation

No.	Topic	✓
65	Requirement Analysis	■
66	Test Scenario	■
67	Test Condition	■
68	Test Case	■
69	Test Data	■
70	Test Steps	■
71	Expected Result	■
72	Actual Result	■
73	Preconditions	■
74	Postconditions	■
75	Test Suite	■
76	Test Plan	■
77	Test Strategy	■
78	RTM — Requirement Traceability Matrix	■
79	Test Summary Report	■

LEVEL 7 — Bug / Defect Management

No.	Topic	✓
80	Bug kya hai?	■
81	Defect kya hai?	■
82	Error vs Bug vs Defect vs Failure	■
83	Bug Life Cycle	■
84	New	■
85	Assigned	■
86	Open	■
87	Fixed	■
88	Retest	■
89	Reopen	■
90	Closed	■
91	Rejected	■
92	Duplicate	■
93	Deferred	■
94	Severity	■
95	Priority	■
96	Severity vs Priority	■
97	Bug Report	■
98	Good Bug Report	■
99	Bad Bug Report	■
100	Bug Tracking	■

LEVEL 8 — Agile & Scrum

No.	Topic	✓
101	Agile kya hai?	■
102	Agile Manifesto	■
103	Scrum kya hai?	■
104	Scrum Team	■
105	Product Owner	■
106	Scrum Master	■
107	Development Team	■
108	Product Backlog	■
109	Sprint Backlog	■
110	User Story	■
111	Acceptance Criteria	■
112	Epic	■
113	Task	■
114	Sprint	■
115	Sprint Planning	■
116	Daily Stand-up	■
117	Sprint Review	■
118	Sprint Retrospective	■
119	Definition of Done	■
120	Velocity	■
121	Story Points	■

LEVEL 9 — Jira

No.	Topic	✓
122	Jira kya hai?	■
123	Jira Project	■
124	Issue	■
125	Bug	■
126	Task	■
127	Story	■
128	Epic	■
129	Creating a Bug	■
130	Bug Assignment	■
131	Priority	■
132	Severity	■
133	Labels	■
134	Attachments	■
135	Comments	■
136	Status	■
137	Workflow	■
138	Sprint	■

No.	Topic	✓
139	Backlog	■
140	Board	■
141	Jira Filters	■
142	Basic JQL	■

LEVEL 10 — Web Application Testing

No.	Topic	✓
143	Website Testing kya hai?	■
144	Login Page Testing	■
145	Registration Page Testing	■
146	Forgot Password Testing	■
147	Search Testing	■
148	Navigation Testing	■
149	Form Testing	■
150	Input Field Testing	■
151	Dropdown Testing	■
152	Checkbox Testing	■
153	Radio Button Testing	■
154	File Upload Testing	■
155	File Download Testing	■
156	Pagination Testing	■
157	Sorting Testing	■
158	Filtering Testing	■
159	URL Testing	■
160	Links Testing	■
161	Image Testing	■
162	Error Message Testing	■
163	Session Testing	■
164	Cookie Testing	■
165	Browser Back/Forward Testing	■
166	Responsive Testing	■
167	Cross-Browser Testing	■

LEVEL 11 — Mobile Testing Basics

No.	Topic	✓
168	Mobile Application Testing	■
169	Android Testing	■
170	iOS Testing	■
171	Installation Testing	■
172	Uninstallation Testing	■
173	Interrupt Testing	■
174	Network Testing	■

No.	Topic	✓
175	Orientation Testing	■
176	Compatibility Testing	■
177	Battery Testing	■
178	Permission Testing	■
179	Notification Testing	■

LEVEL 12 — Database / SQL for QA

No.	Topic	✓
180	Database kya hai?	■
181	DBMS kya hai?	■
182	SQL kya hai?	■
183	Table	■
184	Row	■
185	Column	■
186	Primary Key	■
187	Foreign Key	■
188	SELECT	■
189	WHERE	■
190	ORDER BY	■
191	GROUP BY	■
192	HAVING	■
193	DISTINCT	■
194	COUNT	■
195	SUM	■
196	AVG	■
197	MIN	■
198	MAX	■
199	INSERT	■
200	UPDATE	■
201	DELETE	■
202	JOIN	■
203	INNER JOIN	■
204	LEFT JOIN	■
205	RIGHT JOIN	■
206	Subquery	■
207	Database Testing Basics	■

LEVEL 13 — API Testing

No.	Topic	✓
208	API kya hai?	■
209	API Testing kya hai?	■
210	Client vs Server	■

No.	Topic	✓
211	Request	■
212	Response	■
213	Endpoint	■
214	HTTP	■
215	HTTPS	■
216	GET	■
217	POST	■
218	PUT	■
219	PATCH	■
220	DELETE	■
221	HTTP Status Codes	■
222	200	■
223	201	■
224	400	■
225	401	■
226	403	■
227	404	■
228	500	■
229	Headers	■
230	Parameters	■
231	Request Body	■
232	JSON	■
233	Authentication	■
234	Bearer Token	■
235	API Response Validation	■
236	Postman Basics	■

LEVEL 14 — Python Basics for Automation

No.	Topic	✓
237	Python Introduction	■
238	Variables	■
239	Data Types	■
240	Operators	■
241	Conditions	■
242	if/else	■
243	Loops	■
244	for loop	■
245	while loop	■
246	Functions	■
247	Lists	■
248	Tuples	■
249	Sets	■

No.	Topic	✓
250	Dictionaries	■
251	Strings	■
252	Exception Handling	■
253	Files	■
254	Modules	■
255	Classes & Objects	■
256	OOP Basics	■

LEVEL 15 — Selenium Automation

No.	Topic	✓
257	Automation Testing kya hai?	■
258	Manual vs Automation Testing	■
259	Selenium kya hai?	■
260	Selenium WebDriver	■
261	Browser Automation	■
262	Locators	■
263	ID	■
264	Name	■
265	Class Name	■
266	Tag Name	■
267	Link Text	■
268	Partial Link Text	■
269	CSS Selector	■
270	XPath	■
271	Absolute XPath	■
272	Relative XPath	■
273	Web Elements	■
274	Click	■
275	Send Keys	■
276	Get Text	■
277	Clear	■
278	Dropdown	■
279	Checkbox	■
280	Radio Button	■
281	Alerts	■
282	Frames/iFrames	■
283	Multiple Windows	■
284	Browser Navigation	■
285	Waits	■
286	Implicit Wait	■
287	Explicit Wait	■
288	Fluent Wait	■

No.	Topic	✓
289	Screenshots	■
290	JavaScript Executor	■

LEVEL 16 — pytest / Automation Framework

No.	Topic	✓
291	pytest	■
292	Test Functions	■
293	Assertions	■
294	Fixtures	■
295	Parametrization	■
296	Test Suites	■
297	Test Configuration	■
298	Test Reports	■
299	Page Object Model — POM	■
300	Page Classes	■
301	Base Class	■
302	Utility Classes	■
303	Data-Driven Testing	■
304	Framework Structure	■

LEVEL 17 — Git & GitHub

No.	Topic	✓
305	Git kya hai?	■
306	GitHub kya hai?	■
307	Repository	■
308	Clone	■
309	Add	■
310	Commit	■
311	Push	■
312	Pull	■
313	Branch	■
314	Merge	■
315	Pull Request	■
316	README	■
317	QA Project GitHub par upload karna	■

LEVEL 18 — CI/CD Basics

No.	Topic	✓
318	CI kya hai?	■
319	CD kya hai?	■
320	CI/CD ka purpose	■
321	Jenkins Basics	■

No.	Topic	✓
322	GitHub Actions Basics	■
323	Automated Test Execution	■
324	Test Reports in CI/CD	■

LEVEL 19 — Advanced QA

No.	Topic	✓
325	Risk-Based Testing	■
326	Test Estimation	■
327	Test Metrics	■
328	Defect Metrics	■
329	Test Coverage	■
330	Root Cause Analysis	■
331	Risk Analysis	■
332	Release Testing	■
333	Build Verification	■
334	Production Testing	■
335	Hotfix Testing	■
336	Shift-Left Testing	■
337	Shift-Right Testing	■
338	Continuous Testing	■

LEVEL 20 — Modern QA / AI Testing

No.	Topic	✓
339	AI in Software Testing	■
340	AI-assisted Test Case Generation	■
341	AI-assisted Bug Analysis	■
342	LLM Testing Basics	■
343	Prompt Testing	■
344	Hallucination Testing	■
345	AI Response Evaluation	■
346	AI Application Testing	■
347	Testing AI APIs	■

LEVEL 21 — Real Projects

No.	Topic	✓
348	Project 1 — E-commerce Website Testing	■
349	Login Testing	■
350	Registration Testing	■
351	Search Testing	■
352	Product Testing	■
353	Cart Testing	■
354	Wishlist Testing	■

No.	Topic	✓
355	Checkout Testing	■
356	Payment Flow Testing	■
357	Logout Testing	■
358	Project 1 Deliverables — Test Scenarios	■
359	Project 1 Deliverables — 50+ Test Cases	■
360	Project 1 Deliverables — Bug Reports	■
361	Project 1 Deliverables — Test Execution Report	■
362	Project 2 — API Testing with Postman	■
363	GET/POST/PUT/DELETE Practice	■
364	Authentication Practice	■
365	Response Validation	■
366	Negative API Testing	■
367	Project 3 — Python + Selenium + PyTest	■
368	Automation of Login/Search/Product/Cart/Checkout	■
369	GitHub Project Upload	■

LEVEL 22 — Interview Topics

No.	Topic	✓
370	Tell me about yourself	■
371	What is Software Testing?	■
372	Why Testing?	■
373	SDLC vs STLC	■
374	Verification vs Validation	■
375	Test Scenario vs Test Case	■
376	Severity vs Priority	■
377	Smoke vs Sanity	■
378	Retesting vs Regression	■
379	Functional vs Non-functional	■
380	Bug Life Cycle	■
381	Agile/Scrum	■
382	Jira	■
383	SQL queries	■
384	API basics	■
385	Selenium basics	■
386	Real-world testing questions	■
387	Scenario-based questions	■

Total checklist items: 387

Recommended study method: Fundamentals → video + notes; Test Case/Bug/Jira/SQL/API/Automation → video + hands-on practice; Projects → mostly practical.